

Enhancing Sitting Posture Detection and Alert System with Multi-Sensor Integration

Nor Hazwani Mohamad Nor

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
hazwanimohamadnor@gmail.com

Marlina Yakno*

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
marlinayakno@umpsa.edu.my

Normaniha Abd Ghani

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
normaniha@umpsa.edu.my

Mohd Hisyam Mohd Ariff

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
hisyam@umpsa.edu.my

Norasyikin Fadilah

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
norasyikin@umpsa.edu.my

Zulkifli Musa

Faculty of Electrical and Electronics
Engineering Technology
Universiti Malaysia Pahang Al-Sultan
Abdullah

Pekan, Pahang, Malaysia
zkifli@umpsa.edu.my

Abstract— Poor sitting posture among office workers is a common issue that contributes to musculoskeletal disorders, including low back pain. This paper presents an enhanced sitting posture detection and alert system based on multi-sensor integration. The system incorporates flex sensor for the thoracic spine and MPU6050 sensor for the lumbar spine within a wearable vest to precisely monitor spinal angles. A vibrating motor provides real-time haptic feedback upon detecting poor posture, while data processing is managed by an ESP32 microcontroller, with results displayed on the Blynk application. Performance evaluation, conducted through expert consultation and data collection from ten participants, demonstrated that the multi-sensor system achieved 100% accuracy in detecting good posture, eliminating misclassification errors. In contrast, the single-sensor system attained 80% accuracy, misclassifying 20% of poor postures. Notably, the MPU6050 sensor significantly enhanced detection reliability, particularly in identifying poor posture cases. These findings indicate that the proposed multi-sensor approach improves detection accuracy by 20% over single-sensor systems, offering a promising solution for enhancing posture awareness and reducing posture-related health risks in workplace environments.

Keywords—posture detection, wearable technology, health monitoring, ergonomics, thoracic spine, lumbar spine

I. INTRODUCTION

Modern work life is increasingly characterized by prolonged sitting, particularly among office workers who spend extensive hours in front of computers and digital devices [1, 2]. This sedentary lifestyle has significant adverse effects on the human body, particularly the musculoskeletal system. Proper posture is essential for maintaining the systematic alignment of bones and the balanced distribution of ligament and muscle forces in the trunk. The human spine comprises five regions: cervical, thoracic, lumbar, sacrum, and coccyx, each consisting of independent vertebrae and intervertebral discs (Fig. 1). Maintaining the natural curvature of these spinal regions is crucial for preventing conditions such as kyphosis and low back pain, which are common among individuals who sit for extended periods in improper postures.

Recent studies indicate that low back pain is a growing health concern in Malaysia, with prevalence rates varying across different populations. A 2023 scoping review reported that the prevalence of low back pain ranged from 12.4% to 84.6% among various occupational groups, with nurses (67.9%) and drivers (65.7%) being the most affected [1]. The COVID-19 lockdown further exacerbated the issue, with prevalence rates rising from 64.4% to 83.5% among Malaysian adults. Contributing risk factors include age, gender, occupation, prolonged computer use, poor ergonomics, and increased stress levels [2]. These statistics highlight the urgent need for interventions to mitigate posture-related health risks in professional settings.

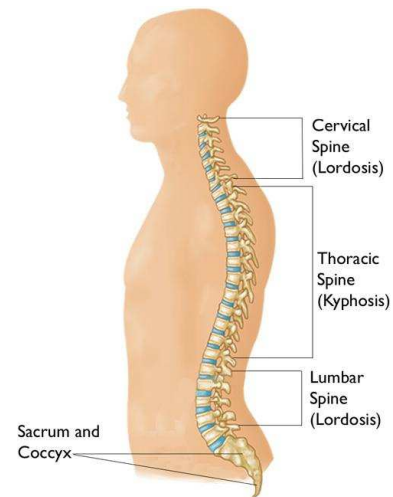


Fig.1: Anatomy of the spine [3]

Poor sitting posture has been linked to several musculoskeletal problems, including back pain, muscle imbalances, transversus abdominis dysfunction, and long-term spinal complications [4], [5]. Office workers are particularly vulnerable due to prolonged sedentary behavior and limited opportunities for movement. Fig. 2 illustrates examples of good and poor sitting postures, demonstrating how individuals often unknowingly deviate from proper spinal alignment during work tasks. Given these challenges, wearable posture monitoring devices have emerged as

promising solutions for posture correction by providing real-time detection and alerts.

To develop an effective posture monitoring system, key considerations include accurate posture detection, user comfort, and practical usability. Various approaches have been explored in the literature, including contact-based sensors such as flex sensors [6], pressure sensors [7], and inertial measurement units (IMUs) [8]. Non-contact wearable solutions, such as elastic straps [9] and vests [10], have also been proposed. These technological advancements offer potential solutions to address posture-related health issues in increasingly sedentary work environments.



Fig.2: Correct sitting posture

This research focuses on enhancing sitting posture detection through multi-sensor integration, combining MPU6050 sensor with flex sensor. The proposed system aims to improve accuracy while ensuring usability in office environments. A vibrating motor provides real-time haptic feedback when poor posture is detected, enabling immediate correction. Additionally, the Blynk application serves as a monitoring interface, allowing users to track their posture patterns and receive detailed analytics over time. This integrated approach addresses the growing need for effective posture monitoring solutions in modern workplaces, where maintaining proper posture remains a significant challenge.

II. RELATED WORKS

Numerous technologies have been developed for posture monitoring and feedback, leveraging various sensor-based wearable devices. Prior research has primarily focused on integrating flex sensors, IMUs, and vibrating motors to assess sitting posture. Bootsman et al. (2019) [11] emphasized lumbar spine monitoring by developing a washable wearable device aimed at preventing occupational back pain. Their system incorporated an IMU sensor placed directly on the lumbar spine and connected to a mobile application for real-time posture feedback. However, its reliance on layered garments raised hygiene concerns, limiting its practicality for daily use. Severin et al. (2020) [12] introduced a head posture recognition system utilizing three inertial sensors (MPU9250 and MPU6050). The system positioned two MPU9250 sensors and one MPU6050 sensor at equal distances of 45 degrees around the neck to collect posture data. While effective for monitoring neck posture, its design was uncomfortable for prolonged use and unsuitable for individuals with hunched back positions. Asadullah et al. (2021) [13] proposed a posture correction system that integrated a flex sensor, an IMU sensor, and an actuator mechanism using fluid-based weight distribution to support spinal alignment. The sensors measured rotational movement and bending angles; however, the excessive material adhesion to the body reduced comfort, making it impractical for office environments. Ijaz et al. (2023) [14] developed a water-driven

spinal corrector utilizing 3D printing technology. The system employed a feedback mechanism comprising a mini-pump, solenoid valve, Arduino Uno microcontroller, and an L298N DC motor driver. By regulating fluid pressure, the actuators applied corrective back forces to promote proper posture. While this system demonstrated effective posture angle reduction, it was unsuitable for sedentary workers due to its bulkiness and required frequent maintenance to manage fluid pressure stability. Overall, while existing posture monitoring solutions demonstrate advancements in sensor-based detection and feedback mechanisms, they often present challenges related to comfort, practicality, and usability in office settings. This study aims to address these limitations by integrating multiple sensors within a lightweight and user-friendly wearable system, offering real-time feedback for effective posture correction.

III. METHODOLOGY

This section details the prototype's body design, circuit connections, overall operational system, and data collection methodology. The prototype was developed to effectively detect poor sitting posture and provide real-time alerts.

A. Flex Sensor for Thoracic Spine Monitoring

In this study, a flex sensor is employed to measure the bending degree of the thoracic spine. The sensor's resistance varies with the bending angle, reaching 100 k Ω at 90°, making it function as a flexible potentiometer. The flex sensor operates with a 10 k Ω resistor in a voltage divider configuration, allowing voltage measurement across the sensor. The voltage divider equation is given by:

$$V_o = V_{cc} \frac{R_{flex}}{R + R_{flex}} \quad (1)$$

where R_{flex} represents the resistance of the flex sensor, R is the fixed resistor, and V_{cc} is the supply voltage. The resistors are connected in series between V_{cc} and ground to enable accurate posture detection.

B. MPU6050 Sensor for Lumbar Spine Monitoring

The MPU6050 sensor is used to measure the roll angle of the lumbar spine region. This IMU integrates 3-axis accelerometer and 3-axis gyroscope, along with an additional temperature sensor. The roll and pitch angles are derived from the accelerometer readings. In the accelerometer readings, ax , ay , and az represent along the x, y, and z axes. The raw accelerometer values are converted to gravity units by dividing by 16,384.2, assuming the sensor is configured with a $\pm 2g$ range. The pitch and roll angles are calculated using the arctangent formulas, and the values are converted to degrees, as shown below:

$$pitch = \arctan\left(\frac{ay}{\sqrt{ax^2 + az^2}}\right) \times \frac{180}{\pi} \quad (2)$$

$$roll = \left(\frac{-ax}{\sqrt{ay^2 + az^2}}\right) \times \frac{180}{\pi} \quad (3)$$

$$yaw = yaw + \left(\frac{gz}{scale\ factor}\right) \times \Delta t \quad (4)$$

where gz represents the gyroscope's z-axis angular velocity. The yaw value tends to drift over time due to gyroscope bias

and noise. To mitigate this issue the following strategies are applied:

- (1) Computing yaw changes over each loop iteration instead of relying on absolute values.
- (2) Performing sensor calibration at startup and subtracting the bias from each subsequent reading to improve accuracy.

C. Hardware Design

The posture correction brace for office workers, illustrated in Fig. 3, is designed to mitigate the effects of non-ergonomic seating conditions. This wearable device is structured like a vest and integrates multiple sensors to monitor spinal alignment and provide real-time feedback for posture correction. The key components of the system include:

- Main board – Positioned on the right side, housing the microcontroller and other essential circuitry.
- Replaceable battery – Located on the left side to ensure easy maintenance and prolonged usage.
- MPU6050 sensor – Placed at the middle lumbar vertebrae to detect lumbar spine angles.
- Flex sensor – Positioned at the middle thoracic vertebrae to measure thoracic spine bending.

This ergonomic design ensures user comfort while accurately detecting postural deviations and delivering corrective feedback.



Fig. 3: Design of the wearable posture detection

D. Circuit Design

The circuit design for the posture detection and alert system, as illustrated in Fig. 4, incorporates the ESP32 microcontroller as its central processing unit, which manages sensor data acquisition and controls the feedback mechanisms. The circuit architecture comprises several critical components and connections. The flex sensor, connected to analog pin D34 of the ESP32, quantifies thoracic spine flexion. Concurrently, the MPU6050 inertial measurement unit communicates via the I2C protocol through pins D21 and D22, providing precise lumbar spine orientation data. For tactile feedback generation, a vibrating motor is implemented on pin D15, activating when postural deviations exceed predetermined thresholds. The circuit design incorporates protective elements, including a diode

positioned across the motor terminals to mitigate inductive kickback phenomena, an NPN transistor functioning as a switching mechanism for efficient motor control, and a resistor to regulate current flow and prevent circuit overload. This integrated design methodology facilitates reliable postural monitoring and timely corrective feedback delivery.

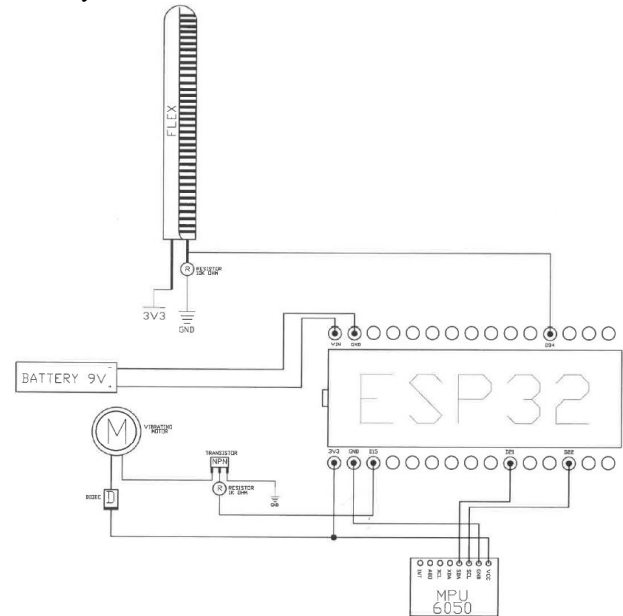


Fig. 4: Circuit design for posture detection and alert system

E. Blynk Application

The Blynk application serves as a comprehensive monitoring and control system for the posture detection and alert system. It provides real-time posture tracking by continuously displaying sensor data, allowing users to monitor their spinal alignment. Additionally, it enables historical data analysis, helping users identify recurring posture issues over time. The application also sends remote notifications when poor posture is detected, ensuring timely corrective action. By integrating these features, the Blynk application offers an intuitive and user-friendly interface for enhancing posture awareness and promoting better sitting habits in workplace environments.

F. System Operational

The block diagram in Fig. 5 illustrates the sitting posture detection and alert system. Designed to monitor and improve sitting posture, this project integrates a flex sensor and an MPU6050 sensor as the primary detecting components. The flex sensor detects bending movement, while the MPU6050 sensor captures orientation and acceleration data. These inputs are processed by the ESP32 microcontroller, which determines posture deviations and triggers corrective actions. When poor posture is detected, the vibrating motor is activated to provide real-time haptic feedback, alerting the user to adjust their sitting position. Additionally, the Blynk application serves as the monitoring interface, displaying posture data and sending notifications to the user's smartphone for continuous tracking and awareness.

The system operates with the ESP32 microcontroller, which manages data acquisition and processing. The flex sensor provides analog input, while the MPU6050 sensor

transmits digital signals via I2C communication. The ESP32 processes these inputs and activates the vibrating motor through GPIO control. IoT connectivity enables data transmission, with the Blynk application displaying real-time and historical posture records. The smartphone application provides an intuitive interface for users to monitor and adjust their posture based on both tactile and visual feedback.

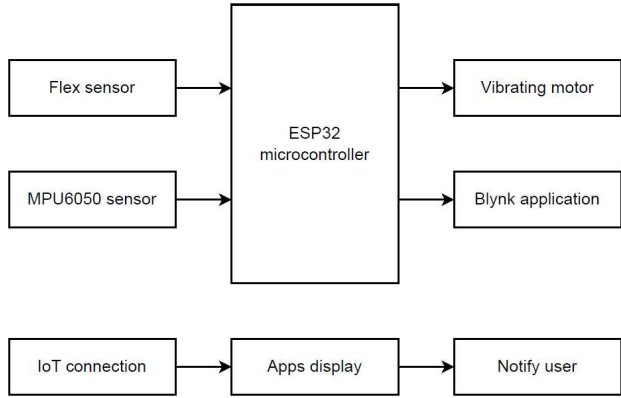


Fig. 5: Block diagram of the posture detection and alert system

Fig. 6 illustrates the system's flowchart, beginning with the calibration of the brace vest posture corrector and the activation of both the flex sensor and the MPU6050 sensor. The system then collects data from both sensors and processes the angle measurements to determine the user's posture. If the flex sensor detects a bending angle greater than 20° and the roll angle from the MPU6050 sensor exceeds 5° , the system classifies the posture as poor [15]. In response, the Blynk application and email notifications display a "Bad Posture" alert. Simultaneously, the system triggers the vibrating motor, which remains active for ten seconds before stopping. If the sensor readings do not exceed these thresholds, the system classifies the posture as proper. The Blynk application and email notifications then display a "Good Posture" message. The system continuously monitors and updates posture data as long as the brace vest posture corrector remains calibrated.

G. Data Posture Collection

Experimental sessions were conducted to collect data from ten participants to determine whether their sitting posture was classified as good or poor. The study took place at the Final Year Student Laboratory, Faculty of Electrical and Electronic Engineering Technology, Universiti Malaysia Pahang Al-Sultan Abdullah, on 11 January 2025. The posture detection assessment procedure while sitting on a chair is shown in Fig. 7. Participants were required to wear the wearable posture detection device comfortably and securely. Both feet had to be flat on the ground (not dangling), and the knees had to be positioned at an angle between 90° and 120° .

In this work, two sensor configurations were tested: (i) a single flex sensor and (ii) a multi-sensor setup combining a flex sensor and an MPU6050 motion sensor. The system was implemented on a microcontroller platform, collecting real-time posture data from users. Posture classification was categorized as either good or poor based on sensor readings. Data were analyzed to compare the accuracy and reliability of both configurations.

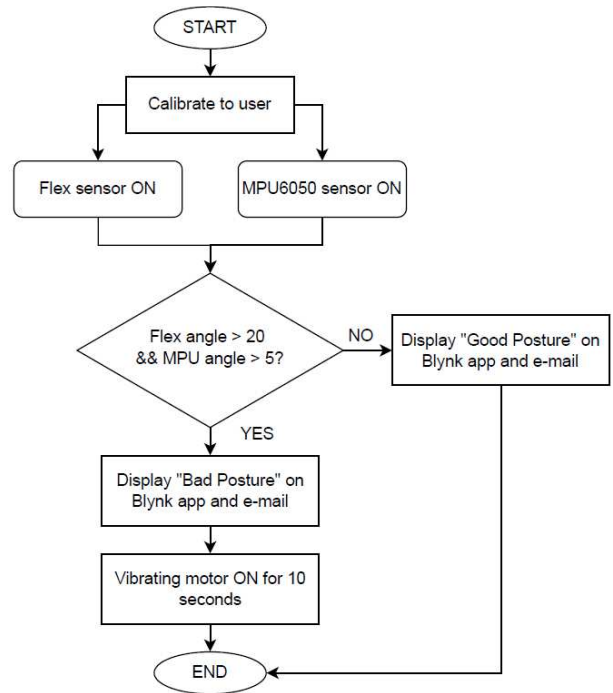


Fig. 6: Flowchart of posture detection and alert system

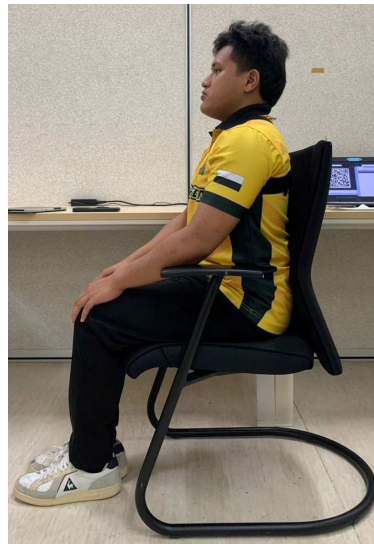


Fig. 7: Sample image of how participant sat on a chair for posture detection

IV. RESULTS AND DISCUSSION

A fully functional wearable posture detection and alert system was designed and implemented, as illustrated in Fig. 8. This system integrates key components for posture monitoring and correction, including an ESP32 microcontroller located on the right side of the brace, a replaceable battery on the left side for extended operation, an MPU6050 inertial measurement unit positioned at the middle lumbar vertebrae to capture lower back alignment, and a flex sensor placed at the middle thoracic vertebrae to monitor upper back curvature.

Table I presents the posture detection results for both the single-sensor and multi-sensor systems, tested on ten participants. The data reveal significant variability in sensor readings across participants, with flex sensor values ranging from 18 to 21 and MPU6050 values from 0 to 8. Notably,

participants classified as having "bad" posture, such as TG22001 and TB21028, exhibited higher MPU6050 readings, suggesting a strong correlation between this sensor's data and posture classification. Conversely, participants with lower MPU6050 readings were generally categorized as having "good" posture.



Fig. 8: A fully functional wearable posture detection and alert system.

TABLE I. COMPARISON RESULTS OF POSTURE DETECTION FOR SINGLE SENSOR AND MULTI-SENSOR INTEGRATION

Participants ID	Single Sensor		Multi-Sensor		
	Flex	Result	Flex	MPU6050	Result
TG21035	20	Good	20	5	Good
TG22001	19	Good	19	8	Bad
TB21025	20	Good	20	1	Good
TB21027	18	Good	18	1	Good
TG21050	19	Good	19	1	Good
TB21013	19	Good	19	1	Good
TB21015	19	Good	19	0	Good
TF21012	21	Bad	21	3	Bad
TB21014	18	Good	18	3	Good
TB21028	19	Good	19	8	Bad

The flex sensor readings displayed less direct correlation with posture classification, indicating that the MPU6050 sensor played a more significant role in the final posture detection outcome. The system appears to use predefined thresholds for both sensors in posture classification; however, these threshold values were not explicitly stated. This variability highlights the potential need for calibration and personalization to account for differences in body structure and sensor placement.

While the multi-sensor integration proved effective, particularly due to the MPU6050's contribution, the small sample size and lack of detailed sensor unit specifications limit the scope of definitive conclusions. Additionally, expert evaluation by a physiotherapist confirmed the system's classification accuracy, showing strong agreement between the system's "good" and "bad" posture detections and visually assessed postural alignment.

Fig. 9 illustrates the posture detection performance of the single-sensor and multi-sensor systems. The results

demonstrate that the multi-sensor system, which integrates a flex sensor and an MPU6050, provides superior posture detection accuracy compared to the single-sensor system, which relies solely on a flex sensor. The multi-sensor system correctly detected all 10 instances of good posture and did not produce any incorrect classifications. In contrast, the single-sensor system accurately detected 8 instances but misclassified 2 cases, leading to detection errors.

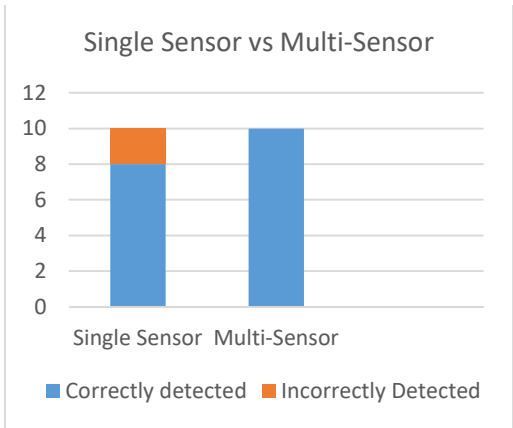


Fig. 9: Comparison of posture detection accuracy between single sensor and multi-sensor system

The improved performance of the multi-sensor system suggests that integrating the MPU6050 enhances detection capabilities by providing additional motion and orientation data. The flex sensor alone may not effectively capture subtle posture variations, which can result in occasional misclassification. The multi-sensor setup significantly reduces false detections, making it a more reliable approach for accurate posture monitoring.

A sample of poor posture for participant ID TB21028 is shown in Fig. 10 (a), illustrating rounded shoulders and a likely forward head posture. In contrast, Fig. 10 (b) presents a sample of good posture for participant ID TB21014, depicting an upright posture with a straight back and aligned shoulders, demonstrating proper ergonomic alignment.

Fig. 11 displays a notification on the Blynk app, indicating the system detected bad posture. This demonstrates the system's ability to wirelessly alert users in real-time, enabling them to correct their posture immediately. The app provides a user-friendly interface to visualize and understand posture alerts.



Fig. 10: A sample of (a) poor body posture and (b) good body posture.

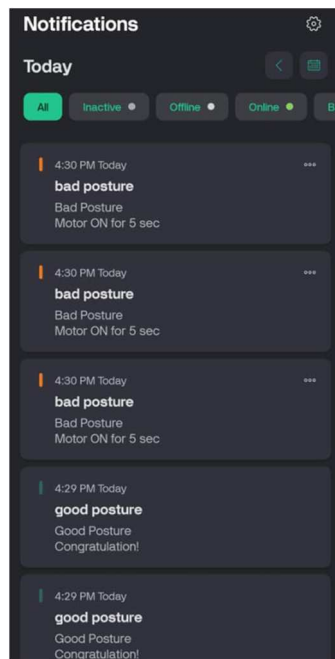


Fig. 11: A sample of notification result display on Blynk app

V. CONCLUSION

The objectives of this research were successfully realized. Firstly, the precision of seated posture detection was significantly enhanced through the integration of an MPU6050 inertial measurement unit with a flex sensor-based system. The incorporation of the MPU6050 sensor at the lumbar spine demonstrably improved both the accuracy and adaptability of the system. Secondly, a cost-effective posture correction vest was developed, integrating essential electronic components and utilizing the Blynk application as a monitoring interface. Thirdly, the performance of the posture and alert system was rigorously evaluated through expert consultation with a physiotherapist. Consequently, the research objectives were effectively met.

For future development, several technical enhancements are recommended despite their associated challenges. Implementation of a rechargeable battery system would require addressing power management and miniaturization constraints. Integration of cervical region sensors would necessitate overcoming calibration complexities and signal processing hurdles. Development of a cloud-based database system for longitudinal tracking would demand solutions for data security and reliable connectivity. Furthermore, a notable limitation of this research was the restricted sample size employed in system evaluation. Subsequent studies should incorporate larger, more diverse participant cohorts across various demographics and postural conditions to establish robust system validation and generalizability.

ACKNOWLEDGMENT

This research was funded by Scheme Final Year Project (FYP), Universiti Malaysia Pahang Al-Sultan Abdullah. Special thanks to Dr. Amalina, physiotherapist at Pusat Kesihatan Universiti Malaysia Pahang Al-Sultan Abdullah, who provided invaluable expertise in posture assessment and critical feedback on the system's design and functionality.

REFERENCES

- [1] A. H. Abas, A. Daud, S. M. Hairon, and M. N. Shafei, "Prevalence and Risk Factors of Low Back Pain in Malaysia: A Scoping Review," *Malaysian J. Med. Sci.*, vol. 30, no. 3, pp. 32–41, 2023.
- [2] C. C. Ying *et al.*, "Impact of Coronavirus Disease 2019 Lockdown on Back Pain Intensity, Prevalence and Associated Risk Factor Among Adults in Malaysia," *Malaysian J. Med. Heal. Sci.*, vol. 19, no. 6, pp. 10–17, 2023.
- [3] W. S. Almuhammadi, E. Agu, J. King, and P. Franklin, "OA-Pain-Sense: Machine Learning Prediction of Hip and Knee Osteoarthritis Pain from IMU Data," *Informatics*, vol. 9, no. 4, pp. 1–28, 2022.
- [4] G. Park and I. Y. Jung, "Real-Time Forward Head Posture Detection and Correction System Utilizing an Inertial Measurement Unit Sensor," *Appl. Sci.*, vol. 14, no. 19, 2024.
- [5] F. Q. S. Dzakupasu *et al.*, "Musculoskeletal pain and sedentary behaviour in occupational and non-occupational settings: a systematic review with meta-analysis," *Int. J. Behav. Nutr. Phys. Act.*, vol. 18, no. 1, pp. 1–56, 2021.
- [6] W. S. Nimi, E. P. Miriam Joshua, R. Jegan, S. Reshma Immaculate, and F. Amy Fedora, "Human Posture Monitoring using Flex Sensor," *Proc. 8th Int. Conf. Commun. Electron. Syst. ICCES 2023*, no. Icces, pp. 55–58, 2023.
- [7] M. C. Tsai, E. T. H. Chu, and C. R. Lee, "An Automated Sitting Posture Recognition System Utilizing Pressure Sensors," *Sensors*, vol. 23, no. 13, pp. 1–21, 2023.
- [8] H. Y. Tang, S. H. Tan, T. Y. Su, C. J. Chiang, and H. H. Chen, "Upper body posture recognition using inertial sensors and recurrent neural networks," *Appl. Sci.*, vol. 11, no. 24, 2021.
- [9] S. Lyu and K. L. LaBat, "Posture modification effects using soft materials structures," *Int. J. Ind. Ergon.*, vol. 84, no. January, p. 103125, 2021.
- [10] S. M. Cerqueira, A. F. Da Silva, and C. P. Santos, "Smart Vest for Real-Time Postural Biofeedback and Ergonomic Risk Assessment," *IEEE Access*, vol. 8, pp. 107583–107592, 2020.
- [11] R. Bootsman, P. Markopoulos, Q. Qi, Q. Wang, and A. A. Timmermans, "Wearable technology for posture monitoring at the workplace," *Int. J. Hum. Comput. Stud.*, vol. 132, no. May, pp. 99–111, 2019.
- [12] I. C. Severin, "Head posture monitor based on 3 IMU sensors: Consideration toward healthcare application," *2020 8th E-Health Bioeng. Conf. EHB 2020*, pp. 4–7, 2020.
- [13] G. M. Asadullah, N. Sabyrov, M. A. S. Kamal, and M. Hazrat Ali, "Design of a fluid-driven 3D printed spinal posture corrector," *Mater. Today Proc.*, vol. 44, pp. 1555–1559, 2021.
- [14] M. K. Ijaz, K. Shomenov, D. Otegen, E. Shehab, and M. H. Ali, "Design and development of a 3D printed water driven spinal posture corrector," *Int. J. Adv. Manuf. Technol.*, vol. 124, no. 5–6, pp. 1457–1471, 2023.
- [15] Y. L. Chen, Y. C. Chan, and L. P. Zhang, "Postural variabilities associated with the most comfortable sitting postures: A preliminary study," *Healthc.*, vol. 9, no. 12, 2021.